



Gas Pressure Regulator RB 1700 RB 1800; RB 1800F

3170992801

AC

Installation and starting up

Installation

- Check that the maximum inlet pressure is not higher than the maximum allowable pressure of the product.
- Arrange enough room for the serving of the regulator and/or the safety slam-shut valve.

Before installing the product in the piping, the following must be checked:

- the upstream piping must be cleaned from all impurities (sand, welding slag, etc.)
- the product must not be visibly damaged.
- the inlet and outlet chambers of the product must be perfectly clean.

Make sure that the direction of gas flow corresponds to the **arrow** on the valve body.

The regulator must be fitted without undue stress on the valve body. Following torques should not be exceeded:

Thread size	3/4"	1"	1 1/2"
Max. torque	80 Nm	110 Nm	140 Nm

Bolts	M12	M14	M16
Max. torque	35 Nm	55 Nm	85 Nm

Sensing line

RB 1700 / RB 1800 / RB 1800F regulator require external sensing lines for regulator¹ and safety valve² controllers. An isolating valve on the safety shutoff valve sensing line **is not permitted**.

Fig. 1 gives a typical example of installation.

Regulator Start-up

Before start-up, the installation must be checked for leaks.

To start-up the regulator properly, proceed as follows:

- check that the upstream and downstream on/off valves and the vent valve are all closed.
- partially open the upstream on/off valve slowly enough to pass a very small amount of gas;
- reset the safety shutoff valve –

See § Relatching the SSV“.

Note that a safety shutoff valve with over- and under-pressure protection is always closed in absence of gas downstream of the regulator.

- check that the pressure rises slowly; the downstream pressure must stabilize around the set value. If the pressure continues to rise, interrupt the start-up procedure by closing the upstream on/off valve and identify the cause of the malfunction;
- after the outlet pressure has stabilized, open the on/off valve completely;
- then slowly open the downstream on/off valve until the piping is completely filled.

At this point, the pressure regulator is operative.

¹ with exception of RB 1700 – size 3/4"

² with exception of RB 1700

Pressure Regulator Setting

The pressure regulator is usually delivered already set to the specifications indicated in the order sheet.

Whenever the set pressure must be modified, this value must be set within the setting range **W**_{ds} of the spring installed.

Make sure there is a flow demand or open the vent valve downstream of the regulator. The vent valve should be closed after the desired set pressure is obtained.

- **To increase the set pressure:** rotate the adjustment screw (a) clockwise – see Fig 2.

Relatching the SSV – see Fig 3

The safety shut-off valve must be re-latch only after verifying the reasons for the SSV to trip and after having re-establish normal operating conditions.

Before re-latching the safety slam-shut valve, make sure that the downstream valve is closed

- *a* – unscrew cover (bb) to remove it;
- *b* – screw it upside down on the stem (aa) until it comes into contact with the controller top casing; continue to screw it slowly, this causes the valve to open gently and the pressure to fill the downstream pipe.

On RB 1800 / RB 1800F – this causes the internal by-pass to open. When the outlet pressure has stabilized,

- *c* – pull the stem and cover outwards to re-latch the trip mechanism. The stem shall remain in “open” position;
- *d* – unscrew the cover and **replace it in operation position** on the controller top casing.

Setting the SSV

The safety slam-shut valve is usually delivered already set to the specifications indicated in the order sheet.

Whenever the set value of the trip pressure **p**_{dso} for over-pressure (and/or **p**_{dsu} for under-pressure must be modified, this value must be set within the set range **W**_{dso} (and/or **W**_{dsu}) of the spring installed.

Do not modify the set value when the SSV controller is not under pressure.

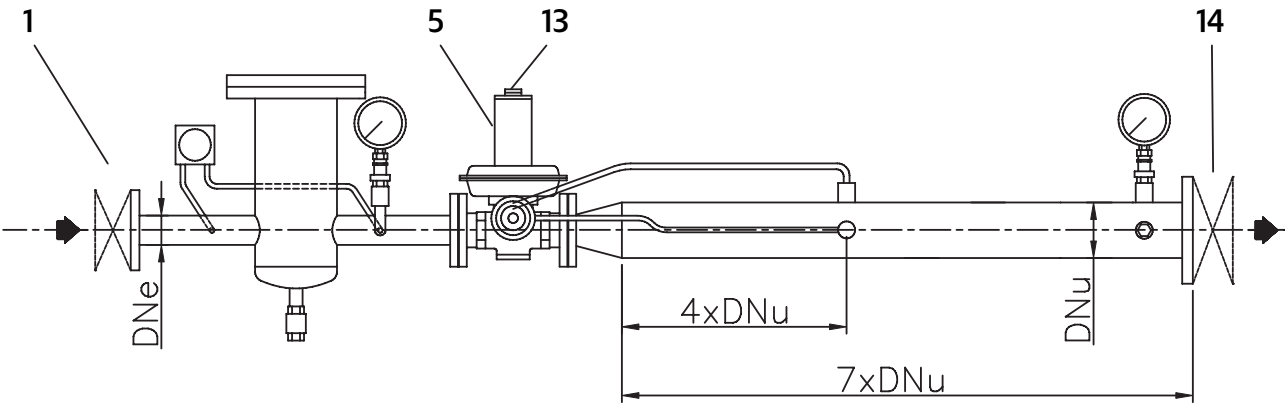
- To increase the set value: rotate clockwise the adjustment screw (b) for upper trip pressure or the adjustment screw (c) for lower trip pressure – see Fig 2.

Table: Connection size of sensing lines

Regulator						
Type	DN	1730- 1731-1732	1720- 1721-1722	1830- 1831-1832	1820- 1821-1822	1810-1 811-1812
RB 1700	3/4"	Ø 2mm	Ø 2mm		-	-
		G 1/2 ¹⁾	G 1/2 ¹⁾			
RB 1700	1"	G 3/4	G 3/4			
RB 1800	1 1/2"	-		G 3/4	G 3/4	G 3/4
SSV						
		1721-1722-1731-1732	1831-1832	1821-1822	1811-1812	
RB 1700	3/4"	Ø 2mm				
		Rp 1/8 ¹⁾				
RB 1700	1"	Ø 2mm				
		Rp 1/8 ¹⁾				
RB 1800	1 1/2"	Rp 1/8	Rp 1/8	Rp 1/8	Rp 1/8	

¹⁾ optional

Fig 1: Installation schematic



1	Inlet valve
5	RB 1700 / RB 1800 / RB 1800F
13	Vent valve
14	Outlet valve

Fig 2 : Adjustment screw

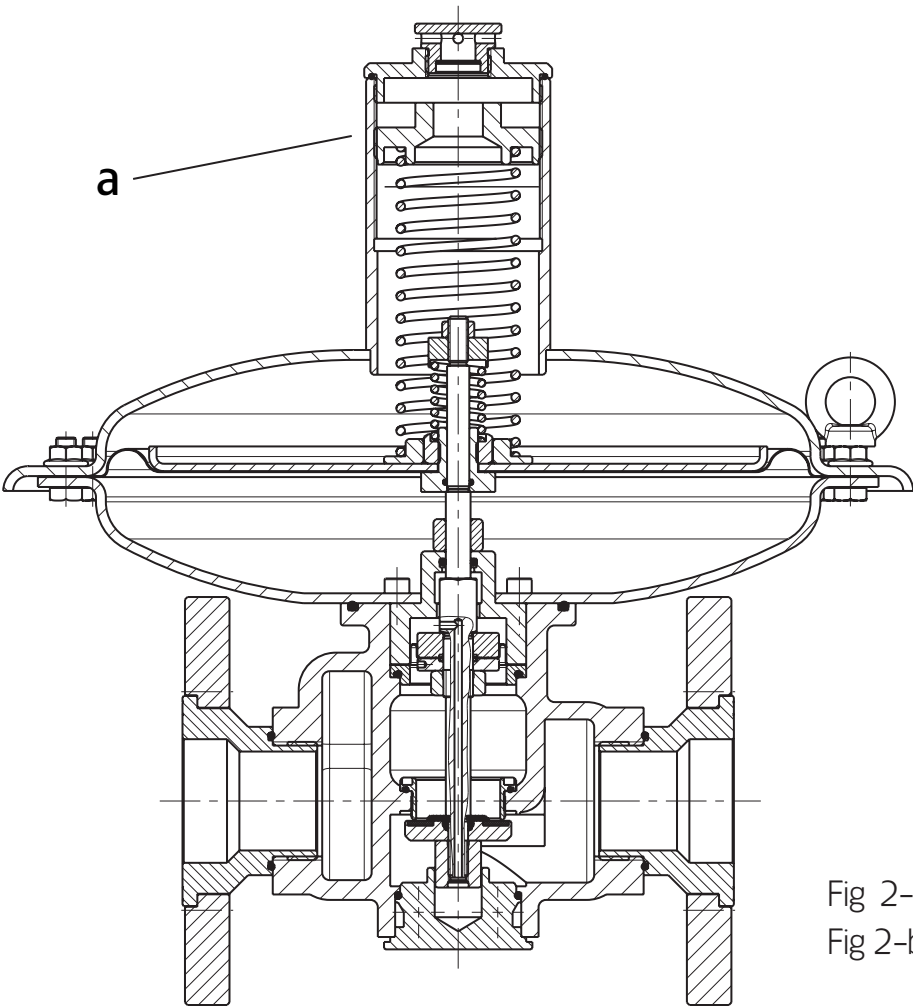


Fig 2-a : Regulator
Fig 2-b: SSV

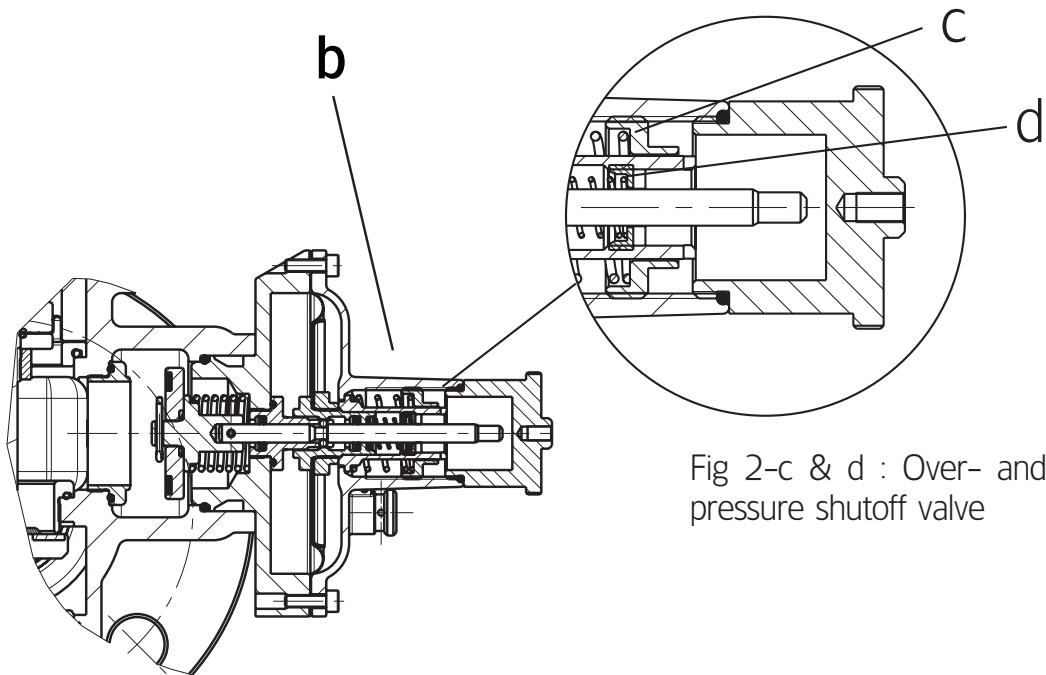


Fig 2-c & d : Over- and under-pressure shutoff valve

Fig 3: SSV resetting

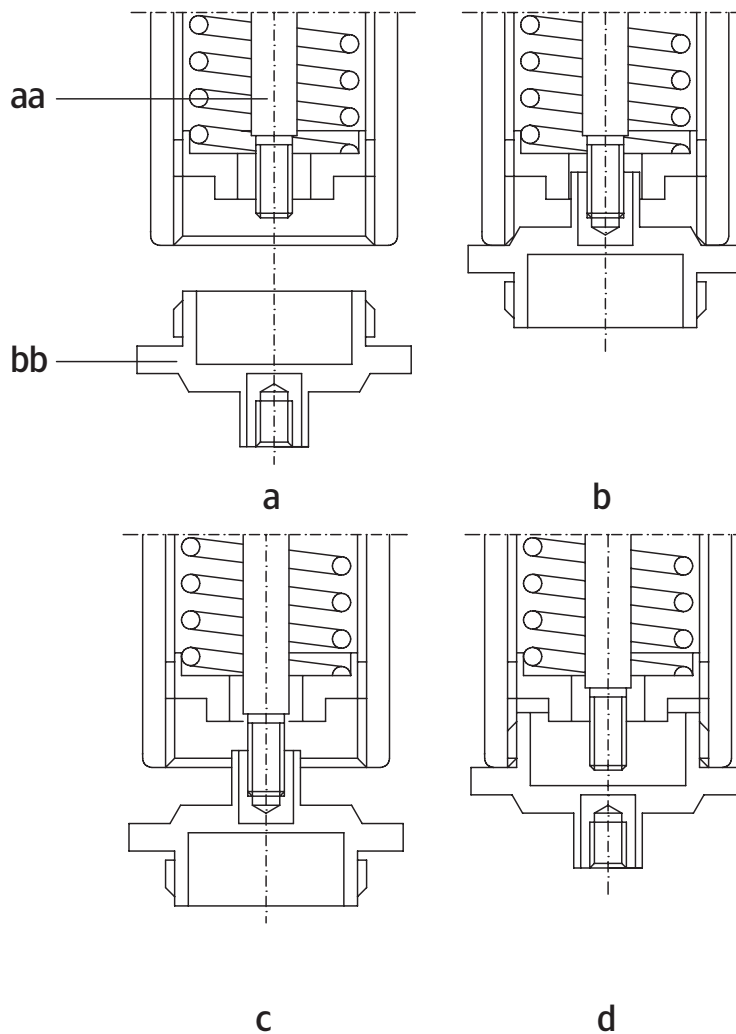


Table1 : Safety valve position sensing

Characteristics	Installation non ATEX	Installation ATEX
Contact power rating	Max. 10 W	Max. 120 mW
Voltage (Break-down)	Max. 180 Vdc	Max. 15 Vdc
Current (Switching)	Max. 50 mA	Max. 50 mA
Capacitance and Inductance	C = 0F, L = 0H	C = 0F, L = 0H
Operation temperature range	-25°C to +55°C	-25°C to +55°C

The device is a simple apparatus as defined in EN 60079-11 §5.7 and is free to be used in ATEX areas.